

Kinematics

Forward and Inverse Kinematics.

FK: angles $\rightarrow$ orientation

IK: orientation $\rightarrow$ angles

Two ways to determine the orientation of end effector.

* RPY Angles

* Euler Angles

RPY

FK: $RPY(\phi_a, \phi_o, \phi_n) = Rot(\alpha, \phi_a) Rot(o, \phi_o) Rot(n, \phi_n)$

通用性
一般公式

$$= \begin{bmatrix} \\ \\ \end{bmatrix}$$

IK: $Rot^T(\alpha, \phi_a) RPY(\phi_a, \phi_o, \phi_n) = Rot(o, \phi_o) Rot(n, \phi_n)$

$$\Rightarrow Rot^T(\alpha, \phi_a) RPY(\phi_a, \phi_o, \phi_n) = Rot(o, \phi_o) Rot(n, \phi_n)$$

$$\Rightarrow \begin{bmatrix} C\phi_a & S\phi_a & 0 & 0 \\ -S\phi_a & C\phi_a & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & 0 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} C\phi_0 & 0 & S\phi_0 & 0 \\ 0 & 1 & 0 & 0 \\ -S\phi_0 & 0 & C\phi_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & C\phi_n - S\phi_n & 0 \\ 0 & S\phi_n & C\phi_n & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} n_x C\phi_0 + n_y S\phi_0 & a_x C\phi_n + a_y S\phi_n & a_x C\phi_n + a_y S\phi_n & 0 \\ -n_x S\phi_0 + n_y C\phi_0 & -a_x S\phi_n + a_y C\phi_n & -a_x S\phi_n + a_y C\phi_n & 0 \\ n_z & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\phi_0 & S\phi_0 S\phi_n & S\phi_0 C\phi_n & 0 \\ 0 & C\phi_n & -S\phi_n & 0 \\ -S\phi_0 & C\phi_0 S\phi_n & C\phi_0 C\phi_n & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \tan \phi_0 = \frac{n_y}{n_x} \quad 2 \text{ values}$$

$$\tan \phi_0 = \frac{-n_z}{n_x C\phi_n + n_y S\phi_n}$$

$$\tan \phi_n = \frac{-a_x S\phi_0 + a_y C\phi_0}{a_x S\phi_0 - a_y C\phi_0}$$

why not use:

$$\sin \phi_0 = -n_z ?$$

Euler

FR $\text{Euler}(\phi, \theta, \psi) = \text{Rot}(x, \phi) \cdot \text{Rot}(y, \theta) \cdot \text{Rot}(z, \psi)$

1k : $\text{Rot}^{-1}(x, \phi) \text{Euler}(\phi, \theta, \psi) = \text{Rot}(y, \theta) \text{Rot}(z, \psi)$

$$\Rightarrow \begin{bmatrix} C\phi & S\phi & 0 & 0 \\ -S\phi & C\phi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} n_x & o_x & a_x & 1 \\ n_y & o_y & a_y & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\theta & 0 & S\theta & 0 \\ 0 & 1 & 0 & 0 \\ -S\theta & 0 & C\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} C\psi & -S\psi & 0 & 0 \\ S\psi & C\psi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} n_x C\phi + n_y S\phi & o_x C\phi + o_y S\phi & a_x C\phi + a_y S\phi & 0 \\ -n_x S\phi + n_y C\phi & -o_x S\phi + o_y C\phi & -a_x S\phi + a_y C\phi & 0 \\ n_z & o_z & a_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} C\theta C\psi & -C\theta S\psi & S\theta & 0 \\ S\psi & C\psi & 0 & 0 \\ -S\theta C\psi & -S\theta S\psi & C\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \left\{ \begin{array}{l} \tan \phi = \frac{a_y}{a_x} \\ \tan \theta = \frac{a_x \cos \phi + a_y \sin \phi}{a_z} \\ \tan \psi = \frac{-a_x \sin \phi + a_y \cos \phi}{-a_x \cos \phi + a_y \sin \phi} \end{array} \right.$$